

Mr. Juan Camilo Rendon Atehortua

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PROFESSIONAL SUMMARY:

Physicist Engineer and AI/Software Engineer with experience in mathematical modelling, signal processing, optimization, and algorithm development using Python, MATLAB, and TensorFlow. Currently pursuing a Master of Professional Engineering in Biomedical Engineering at the University of Sydney, with a background in Electrical Engineering, automation, AI, and applied modelling.

KEY SKILLS:

- Algorithm & Modelling
- Mathematical modelling
- Signal processing
- Numerical methods
- Optimisation algorithms
- Software & Implementation
- Python
- MATLAB
- C/C++ fundamentals
- TensorFlow / TensorFlow Lite/ Pytorch

PROFESSIONAL EXPERIENCE

Senior Deep Learning / AI Software Engineer

Wiemspro – Bio Sensors Suit Group SL (Spain) May 2023 – Present | Remote

- Developed AI and signal-processing algorithms for wearable sensor systems used in exercise analysis and biomechanical performance monitoring.
- Designed Python-based pipelines to process accelerometer, gyroscope, quaternion, and sensor-derived timeseries data.
- Built regression and classification models to estimate physical movement parameters such as range of motion, mean velocity, and maximum velocity.
- Applied signal-processing techniques including filtering, FFT analysis, peak detection, windowing, and temporal feature extraction.

Artificial Intelligence Researcher – Computer Vision and Medical Imaging

Universidad Tecnológica de Pereira (Colombia) 2021 – 2023

- Developed deep learning models for medical image analysis, including classification, segmentation, and multitask learning.
- Designed computational pipelines using Python, TensorFlow, Keras, NumPy, Pandas, and Scikit-learn.
- Applied mathematical and statistical methods to evaluate model performance, uncertainty, and generalisation.
- Worked with large-scale biomedical datasets and implemented reproducible training, validation, and testing workflows.
- Developed models combining image-based feature extraction with quantitative performance analysis.
- Contributed to technical writing, research documentation, and scientific communication.

EDUCATION

Master of Professional Engineering – Biomedical Engineering - University of Sydney - Current

Relevant areas: biomedical systems, engineering analysis, modelling, signal processing, data analysis, instrumentation, and computational methods.

Master of Electrical Engineering - **Universidad Tecnológica de Pereira** - 2023

Focus: artificial intelligence, automation, mathematical modelling, signal processing, and computational systems.
Thesis awarded outstanding distinction.

Bachelor of Engineering Physics- **Universidad Tecnológica de Pereira**- 2020

Relevant areas: optics, physics, differential equations, linear algebra, numerical methods, electromagnetism, programming, data analysis, and mathematical modelling.

SELECTED TECHNICAL PROJECTS

Real-Time Sensor Algorithm Development for Wearable Systems

- Developed algorithms to estimate physical movement parameters from inertial sensors.
- Processed accelerometer, gyroscope, and quaternion signals using Python-based pipelines.
- Applied filtering, FFT, temporal segmentation, and feature extraction to identify movement cycles and estimate biomechanical variables.
- Validated algorithms against experimental reference data and refined models based on performance errors.
- Deployed models into lightweight formats for mobile and embedded-oriented inference.

Optical Sensor Data Analysis for Non-Invasive Glucose Estimation

- Worked with LED and photodiode-based optical sensor data for physiological measurement estimation.
- Analysed multi-wavelength optical signals and their relationship with glucose reference values.
- Implemented feature extraction, signal ratio analysis, statistical correlation, and machine learning models.
- Evaluated model performance using regression metrics and experimental validation strategies.
- Gained practical experience with hardware-acquired optical data and sensor variability.

Medical Imaging Deep Learning and Multi-Task Modelling

- Developed convolutional neural network architectures for image classification and segmentation.
- Applied transfer learning, multi-task learning, uncertainty-based loss weighting, and model evaluation.
- Worked with large-scale X-ray datasets and segmentation datasets for biomedical applications.
- Produced quantitative analysis and technical documentation for scientific reporting.

REFERENCES

Available upon request.